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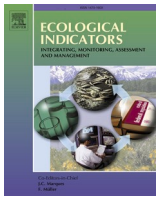

Using LandTrendr to analyze forest disturbance, recovery, and attribution in Hunan province from 2001 to 2024

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ABSTRACT

Forest dynamics monitoring is an important basis for assessing the stability of regional ecosystems and formulating sustainable management strategies, and forest evolution patterns under the interaction of frequent human activities and natural disturbances in subtropical regions have not been fully revealed. This study uses Google Earth Engine and Landsat data (2001–2024) with the LandTrendr algorithm to detect spatiotemporal forest disturbances and recovery in Hunan Province, applying a random forest classifier to identify disturbance types. The results showed that from 2001 to 2024, the forest disturbance area in the study region was 4,723.46 km^2, accounting for 3.48 % of the total forest area; the recovery area was 4,380.33 km^2, accounting for 3.23 %; and the net loss was 343.13 km^2, representing only 0.25 %. Among the disturbance types, weak disturbances accounted for 53.35 % of the total disturbed area, while low-level recovery accounted for 70.4 % of the total recovery area. Fire was the primary disturbance factor, accounting for 40.7 %, followed by land use conversion, which accounted for 27.3 %. This study provides a scientific basis for adaptive management, supporting targeted restoration and policy-making to enhance forest resilience and promote sustainable forest use in subtropical regions.

1. Introduction

Forests are the largest carbon reservoirs within terrestrial ecosystems, playing a critical role in regulating the global carbon cycle (Shi et al., 2024). Dynamic changes in forests directly affect carbon fluxes and storage within forest ecosystems (Dai et al., 2003). Forest disturbances, which refer to events leading to a significant decrease or disappearance in forest coverage and biomass, are the primary drivers of dynamic changes in forest ecosystems. The main forms of disturbances include forest fires, logging, and alterations caused by heavy rains and pest outbreaks (Goward et al., 2008; Huang et al., 2010). These disturbances can be categorized into natural-induced and human-induced types. Natural disturbances, such as fires, and pest infestations, are unpredictable and are often characterized by regional specificity and suddenness, causing direct and notable physical damage to forest ecosystems. Human-induced disturbances, such as deforestation, result from socioeconomic activities and involve planning and control. These long-term impacts may be more profound. Different disturbances have varying ecological and economic effects. (Dale et al., 2001; Lindroth et al., 2009; Mascorro et al., 2015). The FAO (2020) reported that since 1990, the world has lost approximately 420 million hectares of forest, primarily due to deforestation (FAO, 2021). In China's subtropical regions, forest cover has significantly increased, from 12.7 % in 1976 to 22.96 % in 2018 (Shen et al., 2022). This positive trend can be attributed to favorable climatic conditions and the implementation of reforestation policies, which have enhanced the region's forest cover. The region's younger forests and favorable climate make it a potential carbon sink. (Chen et al., 2002; Liu et al., 2012). However, in recent years, extreme

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weather events such as droughts and floods have severely impacted forest ecosystems. Concurrently, human-induced disturbances, such as overlogging and illegal land occupation, threaten sustainable forest use and ecosystem balance. These disturbances will significantly influence the carbon budget of Hunan Province and the subtropical region. (Chen et al., 2002). Given these challenges, efficient methods for detecting forest disturbances are urgently needed to quantify their spatial and temporal characteristics and to better understand their driving forces, thereby supporting effective forest resource management and policy development.

Remote sensing is one of the most effective methods for large-scale forest disturbance detection. For the past 20 years, change detection using remotely sensed imagery has mainly focused on analyzing a few images taken with high temporal resolution, such as MODIS and AVHRR. (Coppin et al., 2004; Banskota et al., 2014). However, it is less capable of detecting small changes in the area. In contrast, the Landsat series has long served as a critical remote sensing data source for detecting forest disturbances over extended time periods, owing to its stability, continuity, and relatively high spectral, temporal, and spatial resolution (Pflugmacher et al., 2014). Several algorithms based on Landsat data, such as LandTrendr, CCDC, VCT, and BFAST, have been developed for detecting forest disturbances. These algorithms differ in temporal frequency, detection metrics, and real-time monitoring capacity (Yin et al., 2022). In cloud-prone subtropical regions, obtaining high-temporal-frequency cloud-free imagery is challenging. The LandTrendr algorithm uses an annual time step and customizable windows for synthesizing cloud-free composites and selecting detection metrics (Wulder et al., 2016; Yin et al., 2022). Implementing LandTrendr on GEE reduces preprocessing time, facilitating large-scale forest disturbance detection (Kennedy et al., 2018). Recent studies using the LandTrendr algorithm have advanced in three main directions: improving detection accuracy through multisource data and spectral indices, analyzing spatiotemporal disturbance patterns, and identifying disturbance drivers. For example, multiseason observations and texture metrics have been shown to significantly enhance detection performance (Qiu et al., 2023). Regional analyses in Hunan provinces have revealed increasing disturbance trends over the past decades (Zhang et al., 2020). In addition, certain studies have focused on classifying disturbance types and uncovering their drivers, highlighting the influence of both natural processes and human activities (Long et al., 2024).

Research has demonstrated that the LandTrendr algorithm is effective in the identification of forest disturbances. Nevertheless, current studies lack refined classification of disturbance types, hindering the identification of different disturbance mechanisms. This limitation hinders the development of forest management and restoration strategies. A detailed classification would improve the precision of forest resource management and ecological protection efforts. This study focused on Hunan Province. Using Google Earth Engine, a comprehensive Landsat dataset for 2001–2024 was constructed. The LandTrendr algorithm has been employed to detect forest disturbance and recovery in Hunan Province, and a detailed spatiotemporal analysis is conducted on the year, intensity, area, and spatial distribution of these events to reveal long-term, large-scale forest disturbance and recovery dynamics. Furthermore, a random forest classification algorithm that integrates multisource features was employed to classify forest disturbances into four categories: logging, fire, building, and conversion (referring to the conversion from forests to grassland, shrub, or cropland). This classification system addresses deficiencies in the existing classification system for forest disturbances in Hunan, aiming to enhance subtropical forests' antidisturbance and recovery capabilities.

2. Study area and data sources

2.1. Study area

Hunan Province (108°47′E ~ 114°15′E, 24°38′N ~ 30°08′N) is located in the southeastern region of China (Fig. 1). The province's topography is predominantly mountainous and hilly, with a subtropical monsoon climate characterized by simultaneous rain and heat and distinct seasonal variations. The average annual temperature of the province is between 16 °C and 18 °C, and the average annual precipitation is between 1,200 and 1,700 mm. As of 2022, the province's forest coverage has reached 59.98 %, with a forest stock of 664 million cubic meters (The Forestry Department of Hunan Province, 2023), making it a key forest area and timber production base in southern China.

2.2. Data acquisition and preprocessing

The monitoring period for forests in Hunan Province covers 2001 to 2024. The remote sensing data consisted of Landsat 5/7/8 TM/ETM+/OLI data with a spatial resolution of 30 m and a revisit cycle of 16 days. Landsat 5 TM (2001–2011, 449 scenes), Landsat 7 ETM+ (2012, 39 scenes), and Landsat 8 OLI (2013–2024, 631 scenes) images were obtained from the GEE cloud platform provided by USGS. For each target year, the study extracts SR Tier 1 datasets, including Landsat 5/7 B1-B5 and B7 bands and Landsat 8 B2-B7 bands. Landsat 5 TM and Landsat 7 ETM + have a 6-bit radiometric resolution, while Landsat 8 OLI has a 12-bit radiometric resolution. A scaling function is applied to improve temporal consistency between them. Clouds and cloud shadows are masked using the QA_PIXEL quality control band. A median composite generates the SR annual composite dataset for the study area from 2001 to 2024.

Land use data were obtained from the Wuhan University 30 m ground cover product (Yang and Huang, 2021), screened to exclude cropland, shrubs, grassland, water bodies, snow, ice, wasteland, impervious surfaces, and wetlands. Forested areas were used as the forest cover base map to reduce the influence of nonforested land.

The MCD64A1 dataset (Artés et al., 2019) of the Google Earth Engine platform provides daily, 16-day, and monthly fire hotspot information. The data had a spatial resolution of 500 m and were resampled to 30 m.

3. Methods

3.1. Technical framework

The following four steps constitute the main components of the research methods and content of this paper (Fig. 2): (1) Construct Landsat time series images; extract forest, cropland, grassland, and shrub from land use data; and resample fire point data for subsequent analysis; (2) Use the LandTrendr algorithm to detect forest disturbance at a pixel level; (3) Use the random forest algorithm to classify extracted patch types; (4) Analyze spatiotemporal pattern and area statistics of forest disturbance and recovery, and calculate disturbance attribution areas..

3.2. Pixel-based forest disturbance detection

The LandTrendr algorithm, developed by Oregon State University in collaboration with the U.S. Department of Agriculture Forest Service (USDA), is a time series segmentation algorithm designed to capture abrupt and gradual disturbances in time series (Kennedy et al., 2007). Time series segmentation is achieved by segmenting the trajectory of image values to detect forest disturbance and recovery (Fig. 3). The NBR (normalized burn ratio) index is sensitive to the chlorophyll content of leaves, charcoal ash content, and soil moisture content, and thus it is extensively utilized for forest disturbance detection (Kennedy et al., 2010, 2015; Zhu et al., 2012). Thus, NBR was selected as the detection index in this study. The NBR is calculated as:

$$NBR = \frac{NIR - SWIR2}{NIR + SWIR2} \quad (1)$$

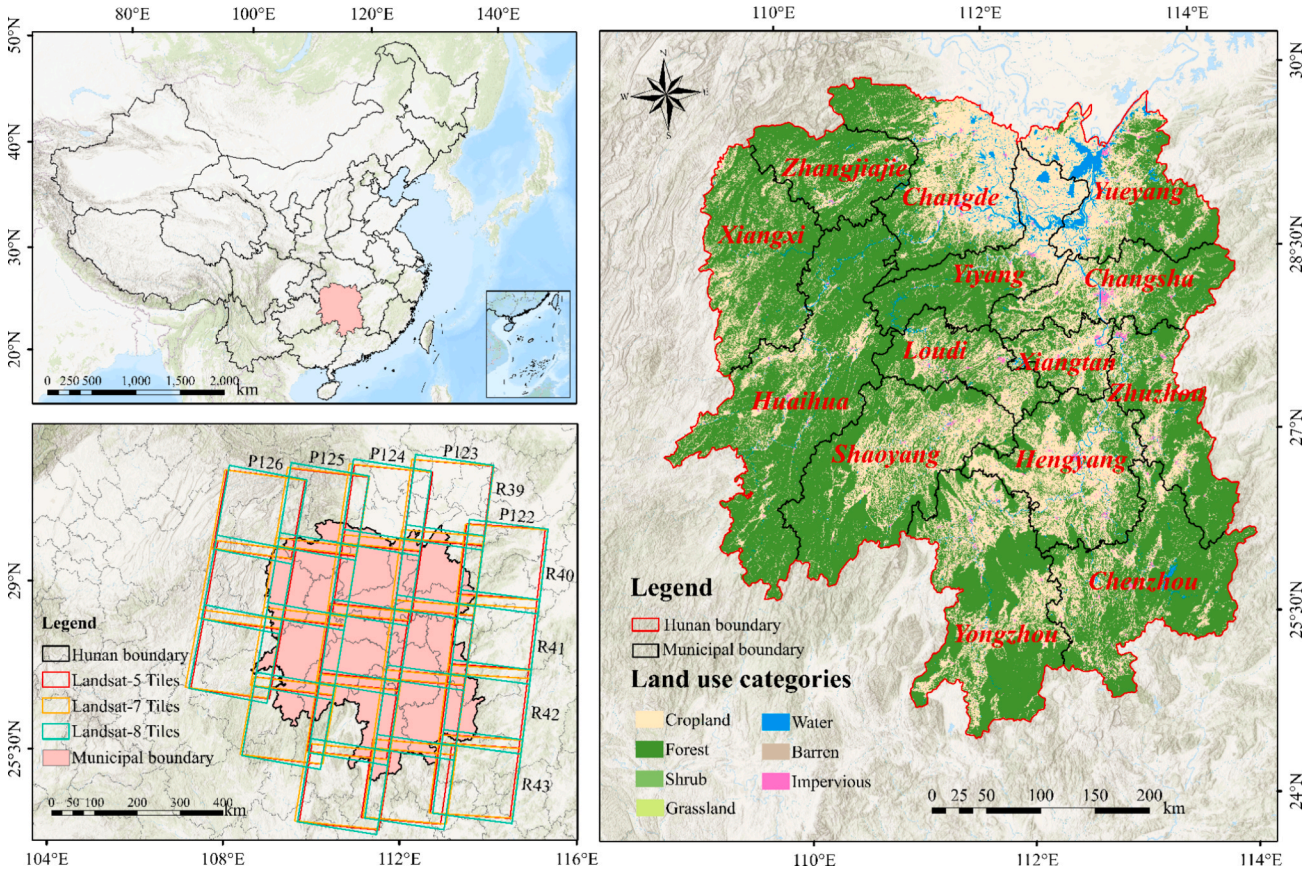

Fig. 1. Location of the study area.

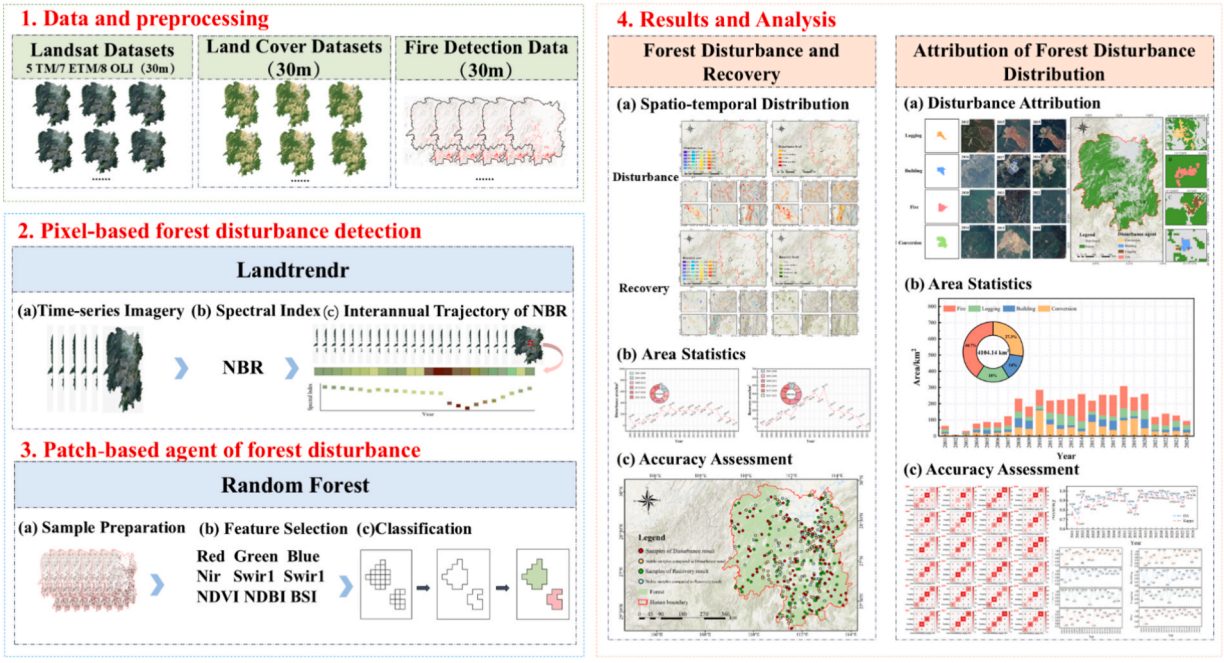

Fig. 2. Technical flowchart.

where NIR and SWIR2 are the near-infrared bands and the second shortwave infrared band of the Landsat image, respectively.

The parameters of the LandTrendr algorithm were adjusted based on the forest characteristics of Hunan's subtropical region and preliminary test results. Specifically, we used: *Max Segments = 6, Spike Threshold = 0.9, Vertex Count Overshoot = 3, Prevent One Year Recovery (Boolean) = true, Recovery Threshold = 0.25, Pval Threshold = 0.05, Best Model Proportion = 0.75,* and Min Observations = 6. A sensitivity analysis was

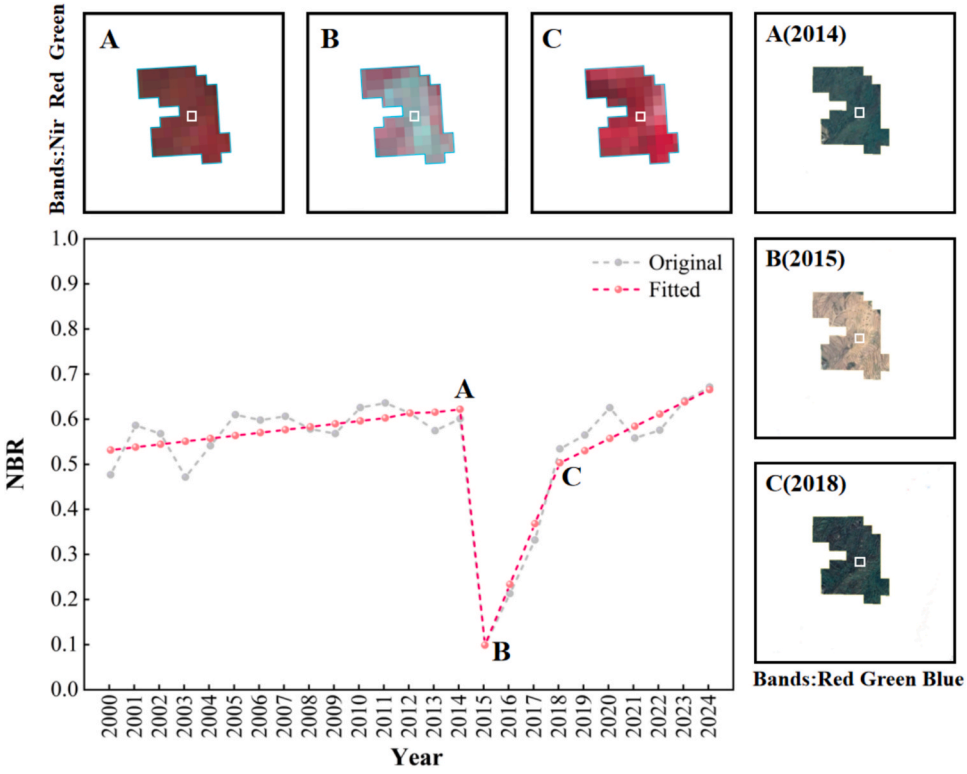

Fig. 3. Trajectory of forest disturbance NBR index based on LandTrendr.

conducted to evaluate the robustness of this configuration, and the results showed that the parameter set is stable and suitable for our study region.

3.3. Patch-based attribution of forest disturbance

Anthropogenic and natural forest disturbances typically occur over large areas (patches) (Kennedy et al., 2015). To reduce the effect of small disturbances, adjacent disturbed pixels in the same year are grouped into patches, excluding those < 0.01 km^2 (Fig. 4). Attribution is only applied to patches larger than 0.01 km^2.

In this study, the random forest algorithm was employed to classify and attribute forest disturbances. As a nonparametric machine learning method, it is robust to a large number of predictor variables (Breiman, 2001). By integrating multiple decision trees, random forest effectively enhances both feature selection and classification performance (Long et al., 2024). The input features for random forest classification include pre- and postdisturbance Landsat images, NDVI, BSI, NDBI, texture features, and disturbance intensity. A total of 10,003 samples were selected using field data and high-resolution Google Earth images from 2001 to 2024 (conversion: 2,495; building: 2,449; fire: 2,499; logging: 2,560). After removing 57 invalid samples, 9,946 valid samples were used, with 70 % for training and 30 % for validation. The formulas for NDVI, BSI, and NDBI are:

$$NDVI = \frac{NIR - RED}{NIR + RED} \tag{2}$$

$$BSI = \frac{(SWIR1 + RED) - (NIR + BLUE)}{(SWIR1 + RED) + (NIR + BLUE)} \tag{3}$$

$$NDBI = \frac{SWIR1 - NIR}{SWIR1 + NIR} \tag{4}$$

where NIR, RED, BLUE and SWIR1 are the near-infrared band, red band, blue band and first shortwave infrared band of the Landsat image, respectively.

The parameterization of the random forest algorithm is detailed in the supplementary material.

3.4. Accuracy assessment

The overall accuracy (OA), user accuracy (UA), producer accuracy (PA), and kappa coefficient, calculated from the confusion matrix (Table 1), were used to evaluate the accuracy of forest disturbance (recovery) and disturbance attribution. In Table 1, for disturbance

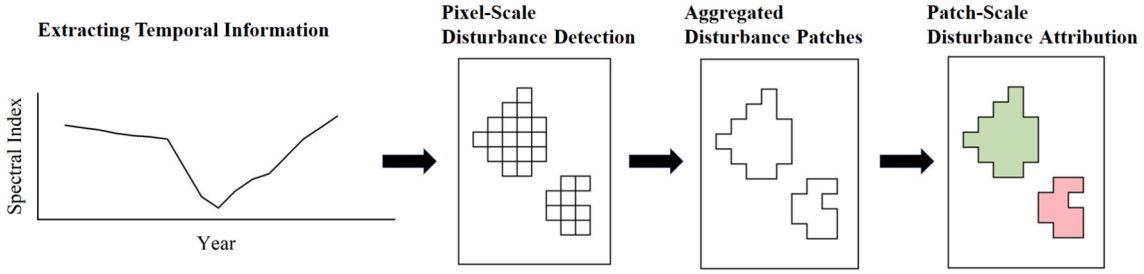

Fig. 4. Forest disturbance attribution method based on random forest.

Table 1
Confusion matrix for forest disturbance (recovery) detection.

	Predicted Negative	Predicted Positive
Actual Positive	False-Negative, FN	True Positive, TP
Actual Negative	True Negative, TN	False-Positive, FP

(recovery), the positive class is disturbed (recovered) forest pixels and the negative class is undisturbed (unrecovered) forest pixels. For attribution accuracy, the positive class is patches correctly identified, and the negative class is misidentified patches. FN is the number of positive samples incorrectly predicted as negative; TP is the number of positive samples correctly predicted; TN is the number of negative samples correctly predicted; FP is the number of negative samples incorrectly predicted as positive.

(1) Overall precision: the proportion of correctly predicted samples among all samples, calculated as:

$$OA = \frac{TP + TN}{TP + FN + FP + TN} \tag{5}$$

(2) User accuracy: the proportion of correctly predicted positive samples among those classified as positive:

$$UA = \frac{TP}{TP + FP} \tag{6}$$

(3) Producer accuracy: the proportion of correctly predicted positive samples among all actual positives:

$$PA = \frac{TP}{TP + FN} \tag{7}$$

(4) Kappa coefficient: measures the nonrandom agreement between classification results and actual observations, with values ranging from −1 to 1. Higher values indicate stronger agreement:

$$Kappa = \frac{OA - P_e}{1 - p_e} \tag{8}$$

$$P_e = \frac{(TP + FN)(TN + FP)}{(TP + TN + FP + FN)^2} \tag{9}$$

To support the accuracy assessment, a total of 2,000 sample points were collected using a stratified random sampling strategy. (Fig. 5), including 500 disturbance points, 500 nondisturbance points during the same period, 500 recovery points, and 500 nonrecovery points between 2001 and 2024. Using Landsat imagery and Google Earth historical high-resolution images, each sample point was visually disturbed, recording whether a disturbance (or recovery) occurred and the time when the disturbance (or recovery) occurred. A confusion matrix was generated using these 2000 sample points to assess the accuracy of forest disturbance and recovery in Hunan Province from 2001 to 2024.

4. Results and analysis

4.1. Accuracy assessment

The confusion matrices generated based on 2000 verification sample points extracted from 2001 to 2024 are shown in Table 2 and Table 3. Table 2 shows the results of disturbance detection accuracy, with an overall accuracy (OA) of 0.96 and a kappa coefficient of 0.91. The UA and PA of both disturbed and no disturbed pixels exceed 0.90. Conversely, the accuracy of recovery detection is relatively low. Table 3 shows that the OA for recovery detection is 0.89, the kappa coefficient is 0.77, and the UA and PA of the recovered and unrecovered pixels exceed 0.85.

Forest disturbance attribution using the Random Forest classifier demonstrated stable and robust classification performance over the 24-year period, with annual OA values mostly exceeding 0.91 and kappa values above 0.80. User's and producer's accuracies varied among disturbance types: logging and construction achieved higher accuracies due to distinct spectral features, whereas conversion exhibited lower accuracy likely caused by spectral similarity with vegetation.

For comprehensive confusion matrices, annual accuracy trends, and detailed accuracy metrics by disturbance type, please refer to the Supplementary Materials.

4.2. Spatiotemporal pattern analysis of forest disturbance and recovery

4.2.1. Spatial and temporal analysis of forest disturbance

The disturbance detection results based on the LandTrendr algorithm revealed pronounced spatial and temporal heterogeneity in forest

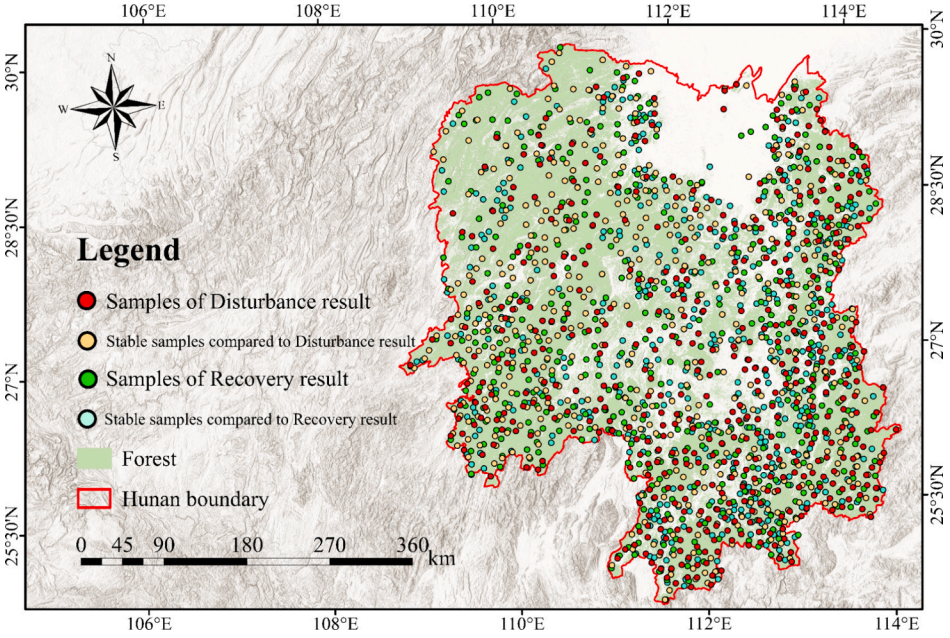

Fig. 5. Distribution map of forest disturbance and recovery validation samples.

Table 2
Pixel-based disturbance detection accuracy results.

Class	Predicted Nondisturbance	Predicted disturbance	Total	UA	PA	OA	Kappa
Actual disturbance	20	480	500	0.95	0.96	0.96	0.91
Actual Nondisturbance	475	25	500	0.96	0.95		

Table 3
Pixel-based recovery detection accuracy results.

Class	Predicted Nonrecovery	Predicted recovery	Total	UA	PA	OA	Kappa
Actual recovery	54	446	500	0.88	0.89	0.89	0.77
Actual Nonrecovery	439	61	500	0.89	0.88		

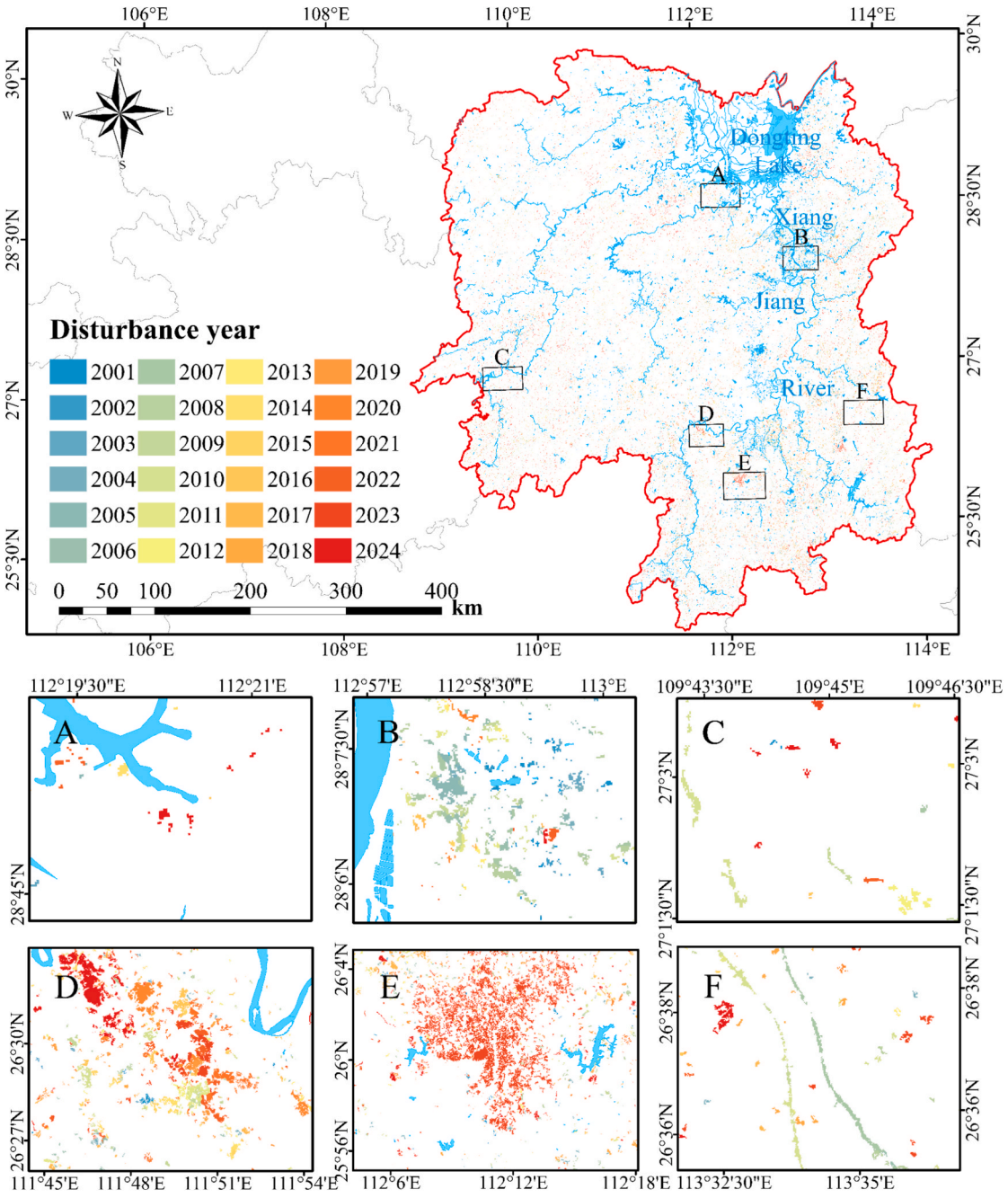

Fig. 6. Spatiotemporal distribution of forest disturbance from 2001 to 2024.

disturbances across Hunan Province from 2001 to 2024 (Fig. 6). Although disturbances were detected annually throughout the study period, the extent and spatial distribution of these disturbances varied significantly from year to year. This study highlights six representative disturbance areas for detailed analysis.

Spatially, there is less forest disturbance in northeastern Hunan, such as in Yueyang and northern Yiyang (Fig. 6-A), which are located in the Dongting Lake basin and are dominated by rice cultivation, with relatively less forest distribution and thus fewer disturbance events. In contrast, disturbances along the Xiangjiang River and its tributaries were mainly concentrated during 2001–2012 (Fig. 6-B), often forming large, contiguous blocks. This pattern may be attributed to early human activities. During this time, rapid economic development in Hunan—particularly urban expansion, agricultural reclamation, and infrastructure construction along river corridors—led to the conversion of substantial forest areas to non-forest land. However, with the implementation of strengthened ecological protection policies, there has been a recent decline in forest disturbance within the Xiangjiang River Basin. The northern and western regions of Hunan exhibit relatively limited disturbances, characterized by a scattered, fragmented distribution. These mountainous areas are sparsely populated, with forest disturbances primarily occurring in proximity to settlements. (Fig. 6-C). In southern Hunan, substantial and widespread forest disturbances were predominantly concentrated during 2019–2024 (Fig. 6-E). As an important forestry economic zone in the province, this region has seen extensive timber harvesting and mineral development activities, frequently resulting in large-scale disturbances. In addition, recent rapid urbanization and infrastructure expansion—including the construction of highways (Fig. 6-F) and industrial parks—have further intensified forest disturbances.

Temporally, disturbances tend to recur in areas previously affected and show strong continuity over time (Fig. 6-D). Previous disturbances often lead to ecological degradation, such as soil erosion, which increases local vulnerability.

The extent of forest disturbance in Hunan Province from 2001 to 2024 is illustrated in Fig. 7. During this period, the cumulative disturbed forest area totaled 4,723.46 km^2, with a peak of 339.29 km^2 in 2018, followed by 319.58 km^2 in 2010. To highlight the periodic characteristics of disturbance dynamics, the study divides the observation period into six four-year intervals. The years 2001–2004 represent a low-disturbance phase, with a cumulative area of 264.87 km^2 (5.6 %), primarily concentrated in 2001 (110.60 km^2) and 2004 (97.79 km^2). In contrast, the 2017–2020 period marks a shift toward high-frequency disturbance, reaching a total of 1182 km^2. In the intermediate periods of 2005–2008, 2009–2012, 2013–2016, and 2021–2024, the disturbance areas were 640.93 km^2 (13.6 %), 1,044.616 km^2 (22.1 %), 1,062.38 km^2 (22.5 %), and 528.47 km^2 (11.2 %), respectively, reflecting a general pattern of increase followed by decline.

4.2.2. *Analysis of forest disturbance intensity*

The spatial and temporal patterns of forest disturbance intensity in Hunan Province from 2001 to 2024 are illustrated in Fig. 8. The Jenks natural breaks method was employed to categorize the disturbance levels into five class intervals: Class I (≤281, weak), Class II (282–367, relatively weak), Class III (368–458, moderate), Class IV (459–564, relatively strong), and Class V (565–1694, strong).

To examine regional disturbance patterns, six representative areas were selected. Results show that high-intensity disturbances are primarily distributed near transportation corridors, urban zones, and agricultural regions, indicating the strong influence of human activities. Low-intensity disturbances are mainly found in the mountainous regions of southern (Fig. 8-F) and western Hunan (Fig. 8-C), where forest cover is high, terrain is complex, and infrastructure is sparse. These areas, dominated by natural forests, are strongly protected under ecological policies such as nature reserves and public welfare forests, which contribute to reduced anthropogenic disturbance. In contrast, high-intensity disturbances are concentrated in northeastern Hunan and low mountainous areas (Fig. 8-B, Fig. 8-D). These zones, characterized by gentle topography and economic activity, are more susceptible to pressures from urban expansion, transportation development, agriculture, and forestry operations.

Overall, weak and relatively weak disturbances dominate forest change in Hunan, affecting 2,519.99 km^2 (53.35 %) of the total disturbed area. Moderate and relatively strong disturbances account for 1,068.40 km^2 (22.62 %) and 748.04 km^2 (15.83 %), respectively. Strong disturbances represent only 387.03 km^2 (8.20 %), indicating that most forests in the province have experienced low levels of disturbance over the study period.

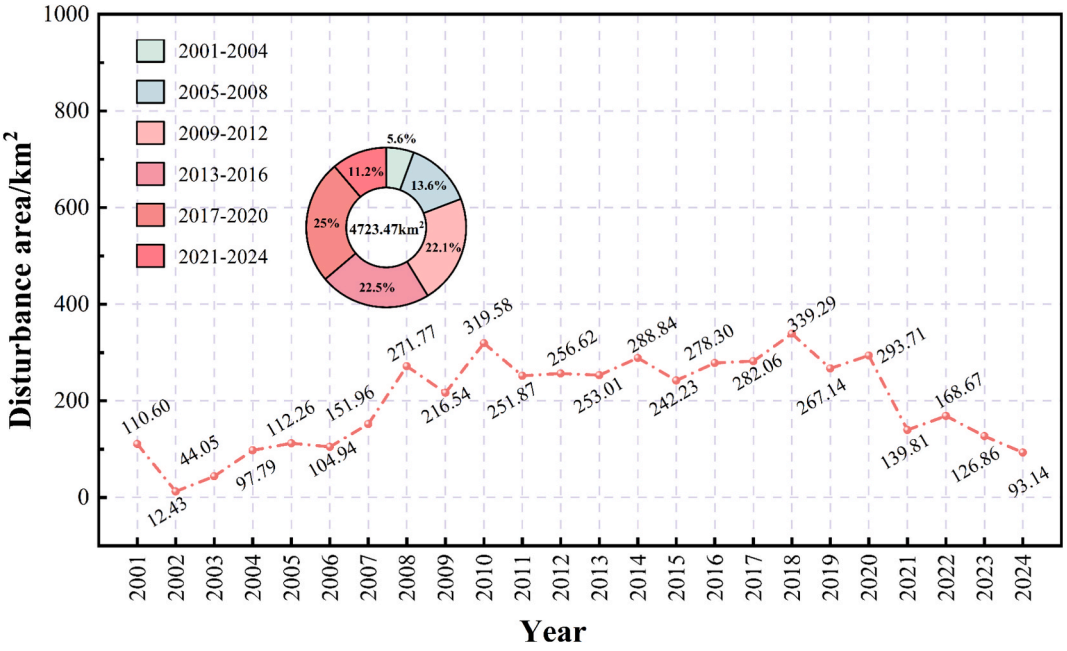

Fig. 7. Forest disturbance area in Hunan province from 2001 to 2024.

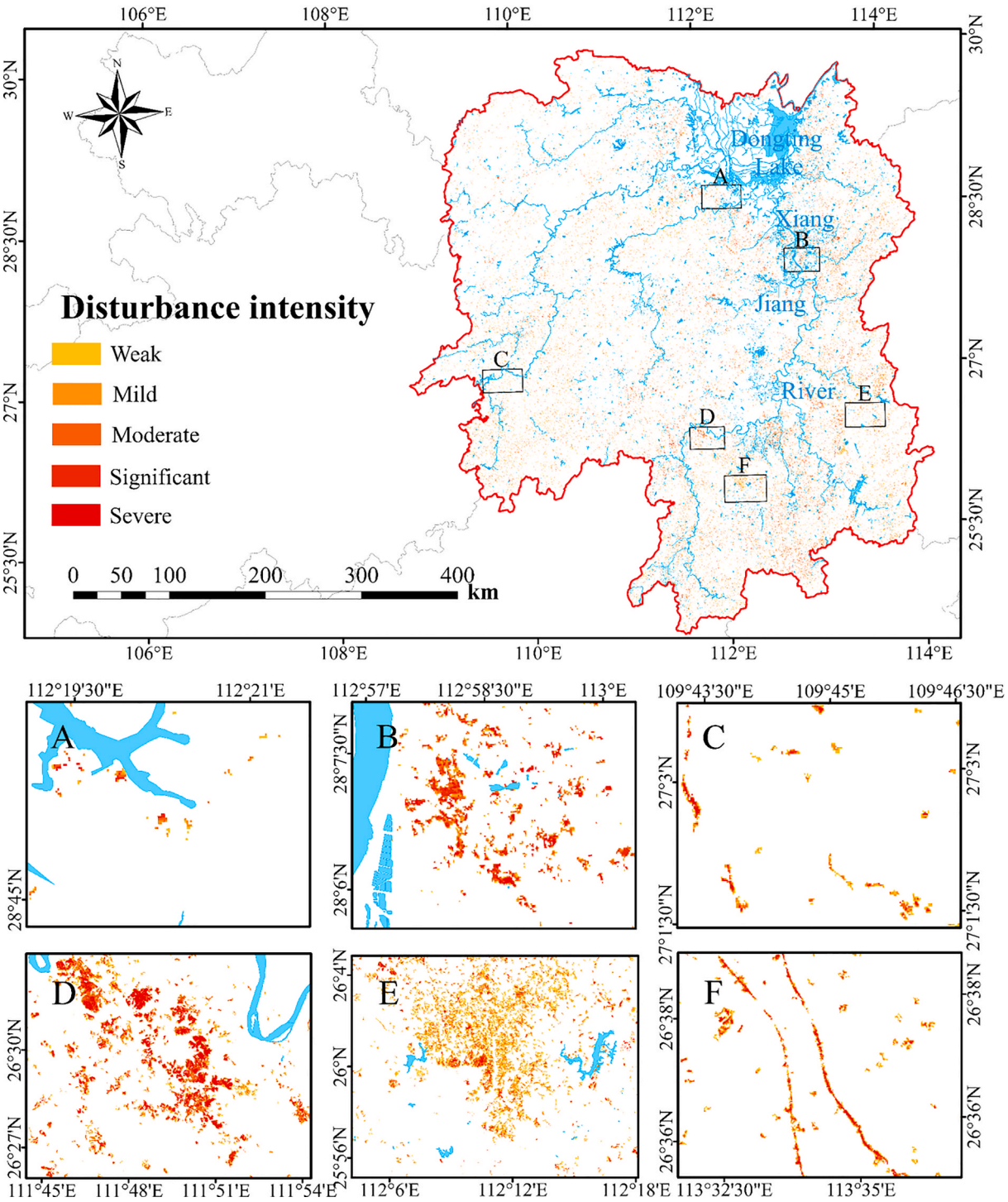

Fig. 8. Forest disturbance intensity distribution map of Hunan Province from 2001 to 2024.

4.2.3. Spatial and temporal analysis of forest recovery

The spatial and temporal patterns of forest recovery in Hunan Province from 2001 to 2023 are shown in Fig. 9. Overall, forests in the province have exhibited a continuous recovery trend over the study period. Spatial variation in recovery is influenced by factors such as topography, land use patterns, disturbance intensity, and ecological restoration measures. Areas subjected to short-term, low-intensity disturbances generally show stronger recovery, while regions experiencing extensive surface hardening, prolonged human disturbances, or topographic limitations tend to exhibit weak or even no recovery.

Analysis of six representative regions reveals that land conversion due to urban expansion—particularly involving irreversible surface hardening—often leads to permanent vegetation loss (Fig. 9-B, Fig. 9-F). In contrast, areas experiencing low-level disturbances (such as road maintenance and temporary construction) in traffic corridors (Fig. 9-F) and around construction sites (Fig. 9-B) tend to undergo natural revegetation within 1–2 years after disturbance cessation. In flat terrains, recovery is typically more continuous and spatially concentrated, supported by coordinated land use and restoration efforts (Fig. 9-D). In contrast, in mountainous areas with steep relief, forest recovery is often fragmented and limited by terrain and soil conditions (Fig. 9-C, Fig. 9-E).

The forest recovery area in Hunan Province from 2001 to 2023 is

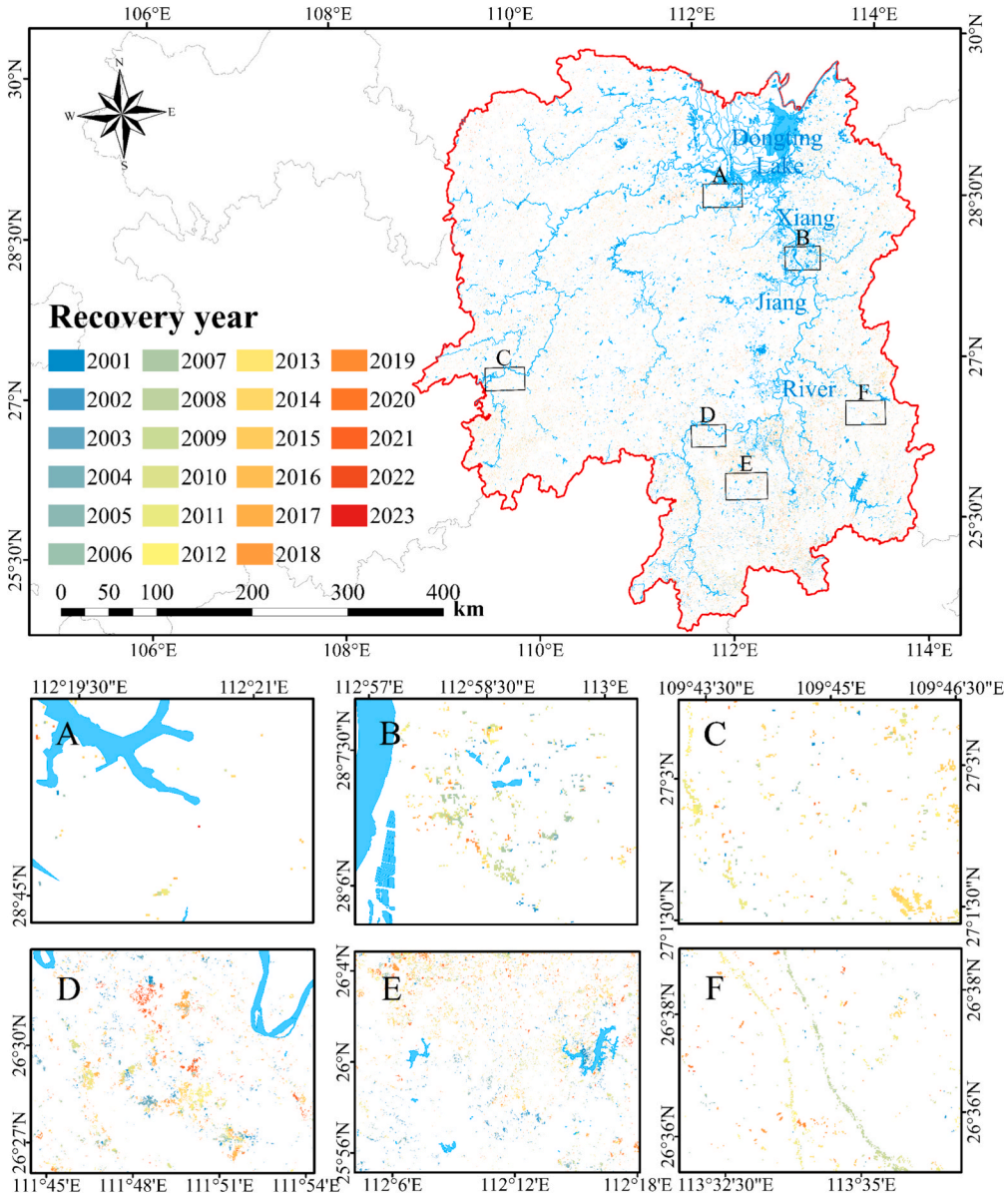

Fig. 9. Spatiotemporal distribution of forest recovery in Hunan Province from 2001 to 2023.

illustrated in Fig. 10. The cumulative forest recovery area reached 4,380.33 km^2 during this period, exhibiting distinct phases characterized by initial growth followed by a gradual decline. The early recovery phase coincided with the initial implementation of the Grain for Green Program (2001–2004). During this period, policy frameworks were undergoing multi-stakeholder coordination and adaptive refinement. As a result, the average annual forest recovery was comparatively low at 64.78 km^2.

After 2005, a growth phase began, with the annual average increasing to 165.28 km^2. After national ecological strategies were adjusted and local assessment mechanisms were strengthened in 2007, forest recovery entered a stable growth phase (2009–2016), with the annual average surging to 346.41 km^2. The peak recovery occurred in 2014, reaching 463.72 km^2. Between 2009 and 2012, the cumulative recovery totaled 1,385.64 km^2, averaging 346.41 km^2 per year.

However, after 2017, rapid urbanization intensified land use competition, thereby slowing the pace of forest recovery. From 2017 to 2020, total recovery amounted to 683.76 km^2, averaging 170.94 km^2 annually. From 2021 to 2023, recovery further declined to just 147.93 km^2 in total, averaging only 49.31 km^2 per year. This decline indicates a policy shift from large-scale expansion to more refined ecological management under the Grain for Green initiative.

4.2.4. *Analysis of forest recovery level*

The forest recovery level in Hunan Province from 2001 to 2023 is shown in Fig. 11. Recovery levels were classified into five categories using the Jenks natural breaks method: Level I (≤356, low), Level II (357–426, low–medium), Level III (427–515, medium), Level IV (516–638, medium–high), and Level V (639–1858, high). The six representative regions exhibit a wide range of recovery levels, from low to high.

Higher forest recovery levels were observed in mountainous areas due to their distinct microclimates, which promote natural forest regeneration. In addition, strict protection policies (e.g., nature reserves and ecological public welfare forests) reduce human disturbances, facilitating faster and more stable forest recovery.

Overall, forest recovery in Hunan Province was dominated by low and low–medium levels, totaling 3,082.01 km^2 (70.36 % of the total). Medium and medium–high levels accounted for 768.96 km^2 (17.56 %) and 406.98 km^2 (9.29 %), respectively. High-level recovery was the least

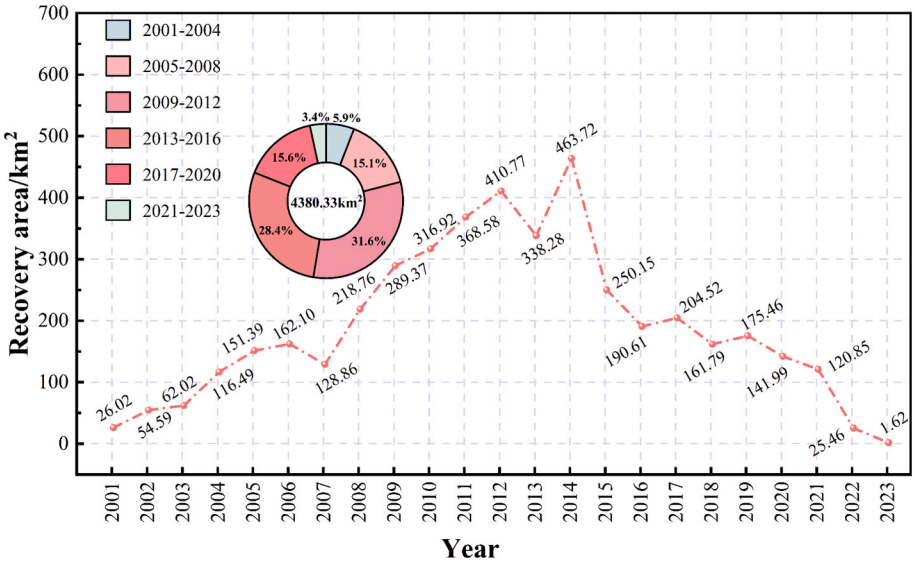

Fig. 10. Forest recovery area in Hunan Province from 2001 to 2023.

common, with only 122.38 km^2 (2.79 %). This suggests that although forests possess a natural capacity for self-recovery, the process is often constrained by factors such as soil degradation and invasive species.

4.3. Analysis of forest disturbance types

A comprehensive analysis of forest disturbance factors in the study area identified four principal categories: logging, fire, construction, and conversion (i.e., transformation of forests into grassland, shrub, or cropland). Random forest algorithm was employed to attribute forest disturbances annually over the 24-year period from 2001 to 2024. Representative disturbance areas for 2015 were selected for visualization, with their spatial distribution and morphological characteristics illustrated in Fig. 12. The results indicate that conversion disturbances display wide, contiguous distribution patterns (Fig. 12-A), typically resulting from large-scale land-use changes (e.g., converting forests to agricultural or urban land). Conversely, fire disturbances are typified by large, continuous patches with a random spatial distribution, wherein patch size and shape are influenced by factors such as fire scale and wind direction (Fig. 12-B). Logging disturbances manifest as small, dispersed patches with irregular boundaries, reflecting local heterogeneity driven by forest management practices (Fig. 12-C). In contrast, construction-related disturbances form regular geometric patterns, as they are constrained by urban planning and infrastructure layouts (Fig. 12-D).

Spatially, forest disturbance types in Hunan Province exhibit marked regional differences. In southern Hunan, logging and conversion disturbances predominate, likely due to abundant forest resources and intense economic activities. Logging is primarily concentrated along forest peripheries for timber production and understory development, whereas conversion disturbances mainly occur in urban–rural interfaces through the transformation of forests into agricultural land. In contrast, eastern and northern Hunan are characterized by a prevalence of construction disturbances driven by rapid urbanization and industrialization, resulting in extensive forest conversion for urban and industrial development. Meanwhile, western Hunan exhibits relatively low disturbance levels, mainly attributable to natural disturbances (e.g., fires) influenced by complex topography and limited human activity.

These regional disparities underscore the need for disturbance-specific forest management and restoration strategies adapted to local socio-ecological conditions.

From 2001 to 2024, forest disturbances in Hunan Province covered a total area of 4104.14 km^2, as shown in Fig. 13. Among the four disturbance types, fire was the most prevalent, followed by conversion, logging, and construction. Specifically, the disturbed forest areas were 1671.91 km^2, 1121.63 km^2, 736.94 km^2, and 573.65 km^2, accounting for 40.7 %, 27.3 %, 18 %, and 14 %, respectively, of the total disturbed area. Further statistical analysis indicated that the area of forest disturbance caused by the four types of disturbance factors during the 12-year period from 2013 to 2024 was greater than that during the 2001–2012 period. The average areas of disturbed forest in the 2001–2012 and 2013–2024 periods were 134.28 km^2 and 207.93 km^2, respectively. The average annual areas disturbed by each factor during these two periods were: Conversion: 40.87 km^2 (2001–2012) vs. 52.60 km^2 (2013–2024); Building: 20.69 km^2 vs. 27.11 km^2; Logging: 24.02 km^2 vs. 37.40 km^2; Fire: 48.50 km^2 vs. 90.83 km^2.

5. Discussion

5.1. Forest disturbance and recovery detection

From 2001 to 2024, forests in Hunan Province exhibited a typical subtropical dynamic characterized by “high fluctuation and low net loss,” with substantial interannual variability in disturbance and a relatively modest overall reduction in forest area. This pattern reflects the frequent alternation between disturbance and recovery in subtropical forest ecosystems, shaped by both natural drivers and anthropogenic activities. The results are consistent with regional findings in central China, where forest dynamics have been strongly influenced by ecological policies and land-use pressures (Zhang et al., 2023).

Spatially, forest disturbances were primarily concentrated in central, southwestern, and northeastern Hunan—areas marked by rapid economic development, intensive infrastructure expansion, and high land-use intensity. Furthermore, long-term implementation of short-rotation logging systems has contributed to recurrent disturbance events in these regions. Conventional harvesting practices remain widespread, intensifying forest structural changes. In contrast, northwestern and south-central Hunan have experienced substantial forest recovery, largely due to sustained ecological initiatives such as the Natural Forest Protection Program and the Grain-for-Green Project, which have been extensively implemented in these key ecological function zones.

In some regions, disturbance and recovery occurred simultaneously, reflecting the influence of compound processes such as post-fire reforestation or the overlap of natural disasters and anthropogenic activities. These patterns underscore the non-linear and multifaceted nature of subtropical forest dynamics, shaped by interactions among forest

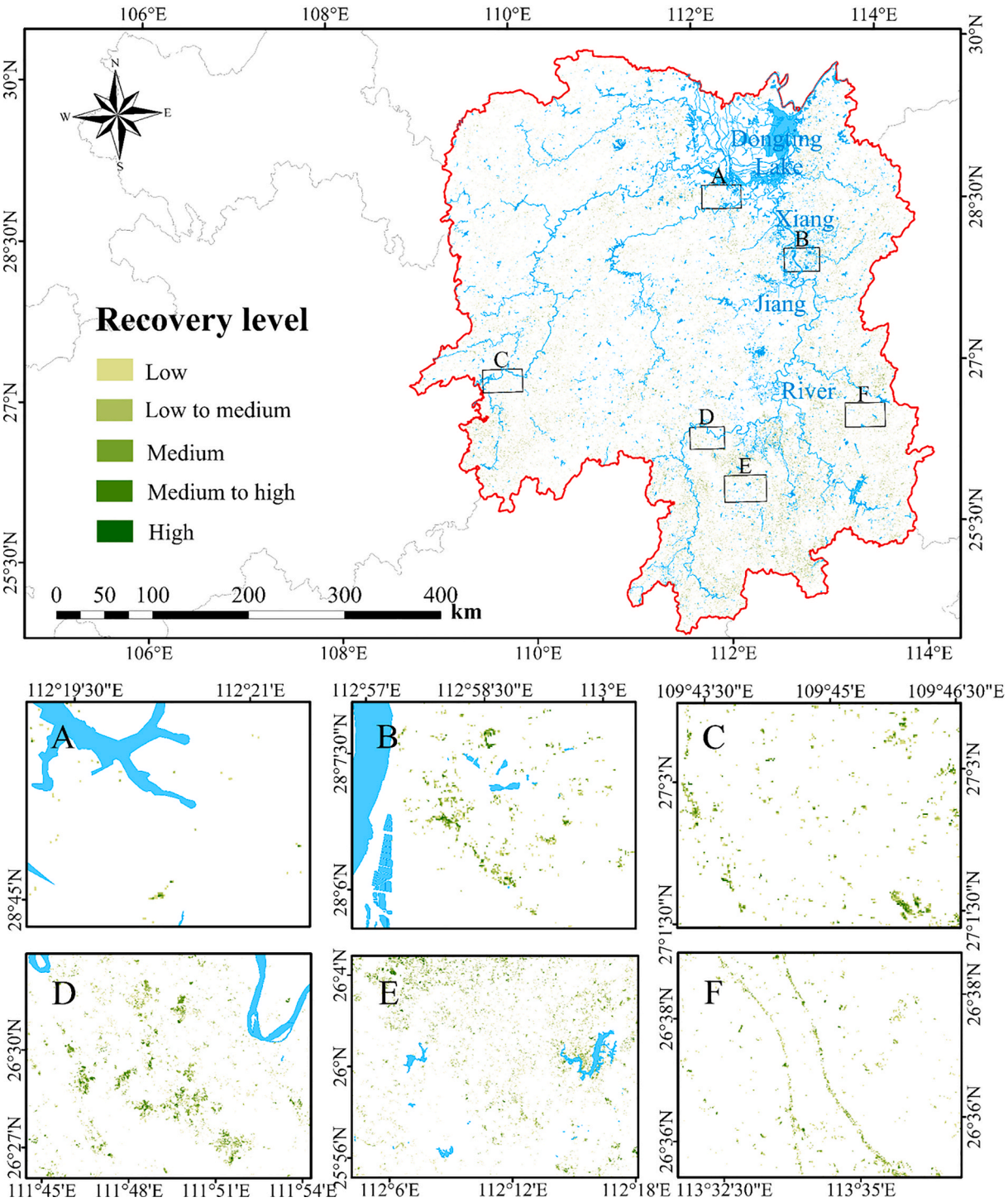

Fig. 11. Distribution of forest recovery levels in Hunan Province from 2001 to 2023.

management regimes, ecological restoration programs, and environmental disturbances.

Given the evident spatial heterogeneity, region-specific forest management strategies are essential. Disturbance-prone areas require stricter regulatory measures and promotion of sustainable harvesting practices, whereas recovery-dominated areas offer opportunities for strengthening reforestation efforts and conserving biodiversity. Future forest policy should account for this spatiotemporal variability to enhance ecosystem resilience and carbon sequestration capacity.

5.2. Forest disturbance attribution analysis

Identifying the drivers of forest change enables a more comprehensive understanding of post-disturbance structural shifts, recovery trajectories, and broader ecological implications (Vogeler et al., 2020).

Disturbance classification reveals that anthropogenic activities are the predominant drivers of forest disturbance in Hunan Province, characterized by long-term and cumulative impacts. Timber harvesting and land conversion, driven by wood demand and agricultural expansion, follow trends consistent with those in other subtropical regions of China (Liu et al., 2018). Construction-related disturbances are closely

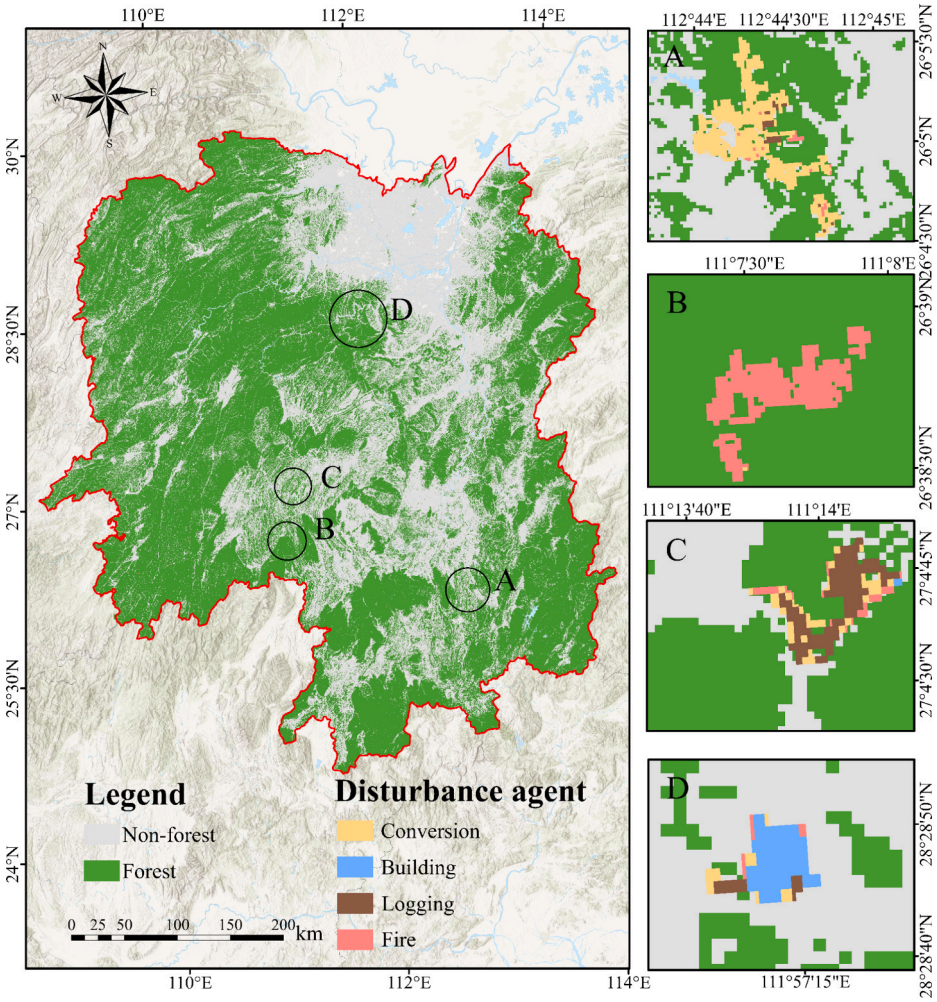

Fig. 12. Forest disturbance attribution mapping of Hunan Province in 2015.

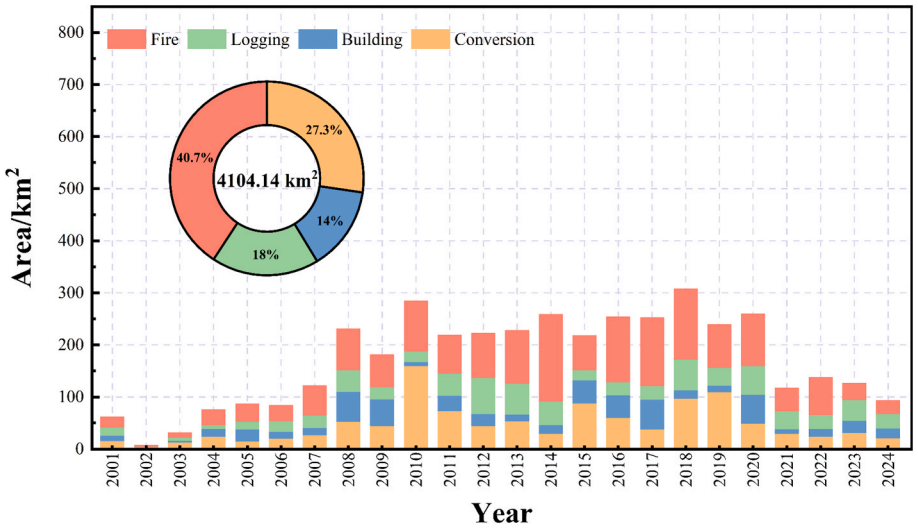

Fig. 13. Disturbance area of four disturbance types in Hunan Province from 2001 to 2024.

linked to regional economic development; for example, eastern and northern Hunan—among the most economically developed areas—undergo intensive land-use transitions and rapid infrastructure expansion. In contrast, ecological civilization and restoration policies have reduced disturbance intensity in parts of western and southern Hunan, highlighting the regulatory role of policy in shaping disturbance patterns (Zhang et al., 2023).

The spatial heterogeneity of these disturbances underscores the need for region-specific forest management strategies. In southern Hunan, frequent but low-intensity harvesting suggests opportunities for

promoting sustainable forest management and ecological compensation to mitigate degradation. In contrast, the high-intensity construction disturbances in eastern and northern Hunan call for integrated land-use planning and stricter enforcement of ecological redlines to balance development and conservation.

Natural disturbances, such as forest fires—often intensified by human activities—require early warning systems, post-disturbance reforestation, and adaptive silviculture to enhance forest resilience. A nuanced understanding of the interactions among human activities, policy frameworks, and ecological processes is essential for building adaptive forest governance in subtropical regions.

5.3. Limitations

This study identified major forest disturbance types in Hunan Province, but challenges remain in distinguishing low-frequency events such as pest outbreaks and ice-snow disasters. These disturbances often exhibit spectral and temporal patterns similar to other disturbance types, complicating their classification. Moreover, the current sample library lacks adequate spatial and temporal coverage of representative cases (e.g., pine wilt disease, canopy collapse from ice storms), limiting the model's generalization and sensitivity to subtle differences.

Future research should establish a multi-source sample acquisition framework by integrating historical disaster records with UAV-based hyperspectral data. Developing a full-cycle sample set covering the onset, progression, and recovery phases can enhance disturbance type differentiation and spatial detection accuracy, providing stronger support for forest monitoring and ecological restoration.

6. Conclusions

This study developed an integrated framework combining Landsat time-series imagery, the LandTrendr algorithm, and random forest classification to achieve high-accuracy detection and attribution of forest disturbance and recovery in Hunan Province from 2001 to 2024. The method effectively captured the spatiotemporal patterns and dominant disturbance types of subtropical forests, demonstrating strong potential for large-scale, long-term forest monitoring.

The findings advance the understanding of forest dynamics and their drivers under subtropical climatic conditions, providing robust technical support for precision management, ecological restoration, and carbon accounting. This work aligns with China's dual-carbon strategy and ecological civilization goals, facilitating the transition from manual inspections to data-driven intelligent forest monitoring and offering a scientific basis for sustainable forest resource management. Future research will focus on integrating higher-resolution satellite data, multi-source data fusion, and advanced deep learning models to further enhance monitoring accuracy and enable broader regional applications.

CRediT authorship contribution statement

Panlin Cao: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Zhuo Zang:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Meng Zhang:** Visualization, Methodology, Investigation. **Xu Wang:** Supervision, Project administration, Funding acquisition. **Xian Tang:** Project administration. **Jiahong Xiang:** Data curation. **Shu Tang:** Software, Methodology. **Jing Wang:** Funding acquisition, Data curation. **Yanan Zhang:** Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecolind.2025.113712.

Data availability

Data will be made available on request.

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